

ENERGY: MAE at each forecast step				
Lookback = 24 steps, Forecast horizon = 72 steps				
Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	160.654	350.367	331.249	318.448
2.000	175.245	255.060	329.889	317.440
3.000	194.791	227.119	326.768	315.508
4.000	211.403	229.810	337.292	327.403
5.000	232.251	242.655	345.552	337.421
6.000	248.565	256.597	355.495	348.279
7.000	259.947	268.813	353.289	344.591
8.000	269.158	278.483	352.609	342.215
9.000	278.155	286.367	362.236	351.764
10.000	286.838	293.363	373.800	363.797
11.000	287.935	299.672	365.416	355.631
12.000	296.051	305.670	386.522	376.411
13.000	305.392	311.034	413.318	402.363
14.000	310.089	315.985	404.585	392.819
15.000	314.894	319.691	396.899	385.922
16.000	317.772	322.596	397.450	388.560
17.000	325.172	325.054	395.349	388.340
18.000	331.282	327.413	392.495	387.095
19.000	334.943	329.253	391.855	387.497
20.000	331.889	331.097	402.633	397.761
21.000	334.876	333.402	399.391	392.515
22.000	334.619	336.468	396.780	386.751
23.000	340.127	340.375	399.065	386.111
24.000	344.415	344.709	412.236	398.291
25.000	347.104	349.607	433.730	419.787
26.000	349.029	353.588	446.982	433.455
27.000	354.094	356.901	440.217	427.908
28.000	354.681	360.193	428.746	418.079
29.000	359.798	363.742	422.349	413.333
30.000	365.709	367.013	435.946	426.857
31.000	369.468	369.582	434.720	423.529
32.000	373.397	371.620	420.333	406.575
33.000	375.886	373.190	412.227	397.715
34.000	382.641	374.579	414.363	399.498
35.000	379.468	375.848	406.655	391.340
36.000	379.824	377.184	419.167	403.160
37.000	384.265	378.470	443.662	425.465
38.000	385.784	379.679	455.095	434.735
39.000	385.182	380.588	453.017	432.834
40.000	385.783	381.362	446.403	427.018
41.000	386.902	382.066	442.356	423.803
42.000	386.996	382.752	432.353	415.800
43.000	387.551	383.304	417.820	402.750
44.000	388.643	384.168	416.008	400.350
45.000	390.644	385.290	416.404	398.516
46.000	390.138	386.537	418.953	399.425
47.000	387.682	388.299	417.229	397.518
48.000	387.468	390.575	421.005	402.288
49.000	389.844	393.120	426.611	408.874
50.000	393.478	394.949	435.074	417.972
51.000	394.683	396.134	438.184	422.184
52.000	396.574	397.165	449.000	434.568
53.000	397.924	398.212	457.588	444.351
54.000	403.715	398.728	460.134	446.612
55.000	400.576	398.880	447.864	431.305
56.000	400.253	398.813	437.842	417.911
57.000	401.453	398.827	439.011	417.041
58.000	403.396	398.881	439.582	417.640
59.000	403.077	398.939	431.488	409.924
60.000	401.811	399.236	435.525	412.839
61.000	401.683	399.383	456.821	432.134
62.000	403.403	399.419	467.341	439.701
63.000	403.462	399.030	464.938	437.257
64.000	402.126	398.200	450.584	427.803
65.000	400.961	397.241	437.265	419.071
66.000	400.148	396.265	430.601	416.378
67.000	401.727	395.189	416.681	405.266
68.000	399.783	394.479	425.255	413.379
69.000	401.314	394.320	437.018	422.622
70.000	401.628	394.366	449.351	433.656
71.000	400.527	394.974	459.749	442.535
72.000	399.182	396.096	468.701	449.935

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. This is the average absolute difference between prediction and truth.